- A multi-scale grid-based noise adaptation mechanism that enhances the performance of diffusion models on low-dimensional datasets.
- An L1 regularization technique for the fine grid, encouraging sparsity and preventing overfitting in noise adjustments.
- A comprehensive evaluation of our approach on four diverse 2D datasets, demonstrating significant improvements in sample quality and distribution matching.
- Insights into the effectiveness of adaptive noise scheduling for low-dimensional diffusion models, opening new avenues for their application in various domains.

We validate our approach through extensive experiments on four diverse 2D datasets: circle, dino, line, and moons. Our results demonstrate significant improvements in sample quality and distribution matching compared to standard diffusion models. We observe KL divergence reductions of up to 36.8% for the line dataset and 22.5% for the moons dataset, indicating a substantial enhancement in the model's ability to capture the underlying data distribution. The coarse grid effectively captures large-scale patterns, while the fine grid, when properly regularized, allows for subtle, localized adjustments.

Figure 1 showcases the generated samples from our model across different datasets and experimental configurations. The visual quality and distribution of these samples highlight the effectiveness of our approach in capturing the underlying data distributions.

The success of our grid-based noise adaptation mechanism in low-dimensional spaces suggests promising directions for future research. Extending this approach to higher-dimensional data and exploring its applicability to specific domain problems, such as financial modeling or geospatial analysis, could lead to significant advancements in these fields. Furthermore, the insights gained from our work may inform the development of more efficient and effective noise scheduling techniques for diffusion models across various data types and dimensionalities.

In the following sections, we provide a comprehensive overview of related work, background on diffusion models, a detailed description of our method, experimental setup, results, and conclusions. Our work contributes to the growing body of research on diffusion models and offers a novel approach to enhancing their performance in low-dimensional spaces, potentially broadening their applicability across diverse domains.

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2 Related Work

Our work on enhancing diffusion models for low-dimensional data builds upon several key areas of research in generative modeling. We discuss relevant advancements in adaptive noise scheduling, applications of diffusion models to low-dimensional data, and spatial adaptations in generative models.

2.1 Adaptive Noise Scheduling in Diffusion Models

Recent work has highlighted the importance of noise scheduling in diffusion models. The Elucidating Diffusion Models (EDM) framework Karras et al. (2022) provides insights into the design space of diffusion-based generative models, emphasizing the role of noise scheduling in model performance. While EDM focuses on high-dimensional data such as images, our work extends the concept of adaptive noise scheduling to low-dimensional spaces.

Unlike EDM, which proposes a global noise schedule optimization, our approach introduces spatially-aware noise adaptation through a multi-scale grid mechanism. This distinction is crucial in low-dimensional settings, where the limited spatial complexity necessitates more fine-grained control over the noise distribution.

2.2 Low-Dimensional Applications of Diffusion Models

The application of diffusion models to low-dimensional data has gained attention recently, with works like TabDDPM Kotelnikov et al. (2022) adapting these models for tabular data generation. While

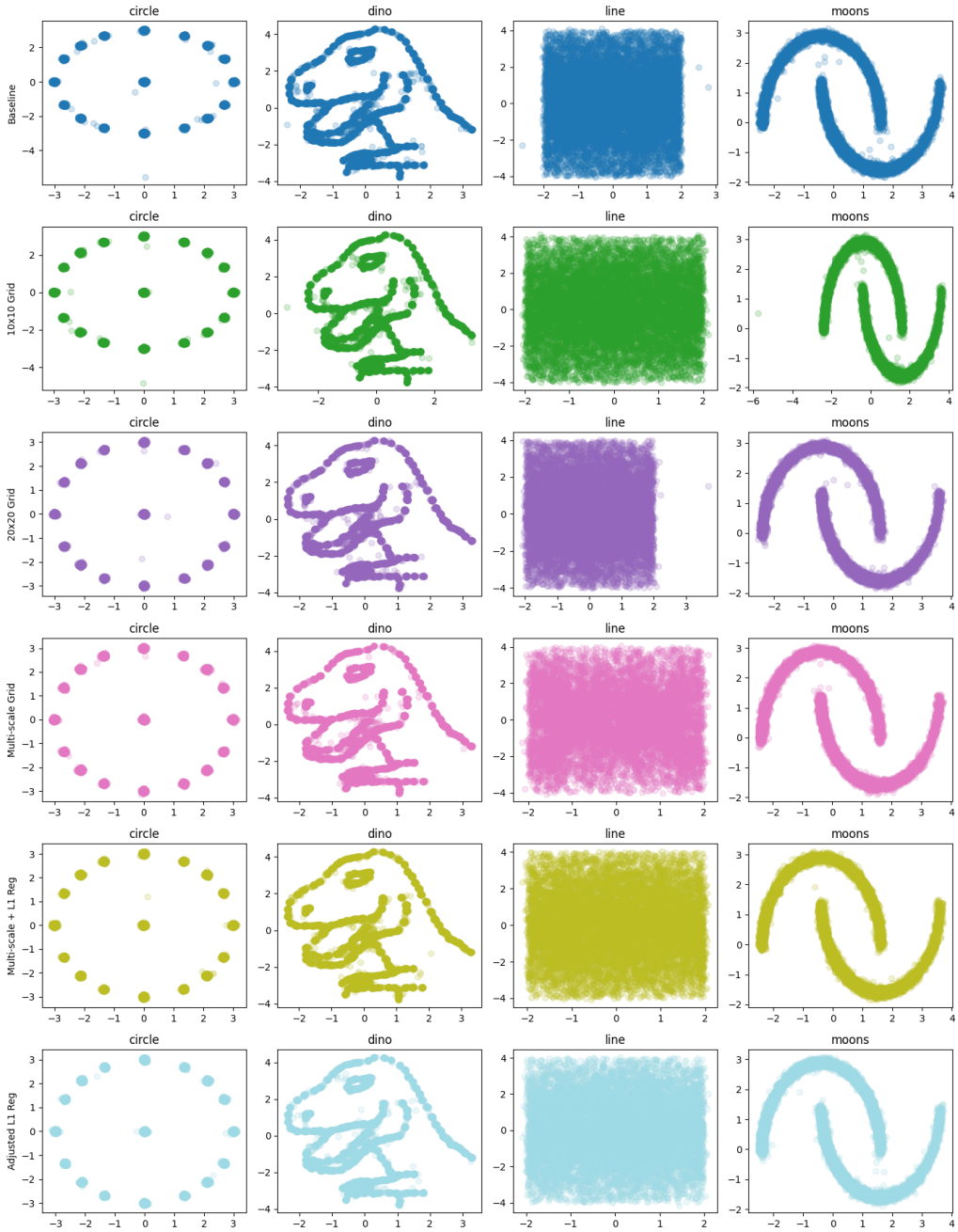


Figure 1: Generated samples from our multi-scale grid-based noise adaptation model for circle, dino, line, and moons datasets across different experimental configurations.

TabDDPM demonstrates the potential of diffusion models in handling structured, low-dimensional data, it primarily focuses on categorical and mixed-type variables.

Our work differs from TabDDPM in several key aspects. First, we specifically target continuous 2D data, which presents unique challenges in capturing spatial relationships. Second, our multi-scale grid approach provides a more flexible framework for adapting to various low-dimensional distributions, as evidenced by our experiments on diverse 2D datasets (circle, dino, line, and moons).

2.3 Grid-Based and Spatial Adaptations in Generative Models

Grid-based and spatial adaptations have been explored in other generative modeling frameworks, particularly in GANs Goodfellow et al. (2014) and VAEs Kingma & Welling (2014). These approaches often involve spatially-aware discriminators or encoders to capture local structures in data.

Our work brings the concept of spatial adaptation to diffusion models, addressing the unique challenges posed by the iterative denoising process. Unlike GANs or VAEs, where spatial adaptations primarily affect the generation or encoding step, our multi-scale grid mechanism influences the entire diffusion trajectory. This allows for more nuanced control over the generation process, particularly beneficial in low-dimensional spaces where small variations can significantly impact the final distribution.

In conclusion, our work addresses a gap in the existing literature by introducing a spatially-aware, multi-scale noise adaptation mechanism specifically designed for low-dimensional diffusion models. By combining insights from adaptive noise scheduling, low-dimensional applications, and spatial adaptations in generative models, we provide a novel approach that enhances the performance of diffusion models in capturing complex low-dimensional distributions.

3 Background

Diffusion models have emerged as a powerful class of generative models, building upon the foundations of variational autoencoders (VAEs) Kingma & Welling (2014) and generative adversarial networks (GANs) Goodfellow et al. (2014). These models are rooted in the principles of non-equilibrium thermodynamics Sohl-Dickstein et al. (2015) and have gained significant attention due to their ability to generate high-quality samples across various domains Ho et al. (2020).

The core concept behind diffusion models is the gradual addition of noise to data, followed by learning to reverse this process. This approach allows the model to capture complex data distributions by breaking down the generation process into a series of simpler denoising steps Yang et al. (2023). The process can be described in two main phases:

1. Forward diffusion: A data point $\mathbf{x}_0$ is gradually corrupted with Gaussian noise over T timesteps, resulting in a sequence of increasingly noisy versions $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T$.

2. Reverse diffusion: The model learns to reverse this process, generating samples by iteratively denoising random noise.

Recent advancements in diffusion models have focused on improving their efficiency and applicability to various data types. Notable works include the Elucidating Diffusion Models (EDM) framework Karras et al. (2022), which provides insights into the design space of diffusion-based generative models, and TabDDPM Kotelnikov et al. (2022), which adapts diffusion models for tabular data generation.

While these advancements have significantly improved the performance of diffusion models in high-dimensional spaces, their application to low-dimensional data presents unique challenges that require careful consideration.

3.1 Problem Setting

Let $\mathcal{X} \subset \mathbb{R}^d$ be a low-dimensional data space, where d is typically small (e.g., $d = 2$ in our experiments). The forward diffusion process is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \tag{1}$$

where β_t is the noise schedule at timestep t, and $\mathcal{N}(\mu, \Sigma)$ denotes a Gaussian distribution with mean μ and covariance matrix Σ.

The goal is to learn a reverse process that can generate high-quality samples by gradually denoising random noise:

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